

RUDDY SIMONPOUR

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 [Portfolio](#)

Recent update: 7/20/2026

WORK EXPERIENCE

Rivo Holdings LLC

San Diego, CA

Data Scientist

March 2025 - Present

Modeling

- Built and deployed a real-time **XGBoost underwriting** model to identify applicant risk, improve default prediction, and determine which cases required analyst intervention versus automated decisioning.
- Improved model precision/recall by **15% through Optuna-based** tuning and iterative feature evaluation, supporting stronger risk segmentation and a reduction in default rate from 15% to 12%.
- Leading an LLM-assisted underwriting pilot by **fine-tuning Mistral-7B with QLoRA** on transaction-level data, customer behavior, and risk signals to support classification and analyst decisioning; benchmarking against current workflows across accuracy, manual-review effort, and edge-case handling.
- Designed Enterprise Management System (EMS)-based controls to centralize model thresholds, eligibility rules, and routing logic, enabling risk and operations teams to adjust decisioning strategies based on model performance, loan outcomes, and macroeconomic trends without redeploying code.

Engineering/Automation

- Converted customer journey patterns into predictive risk features for underwriting models, improving precision/recall by 2–3% per retraining cycle and strengthening model-driven routing between automated decisions and analyst review.
- Built a **model-driven loan decisioning engine** that identifies which applications can be automated versus which require analyst intervention, combining eligibility rules and credit risk scores to support approval, funding, and manual-review routing with 92-95% decision agreement with analyst-reviewed outcomes.
- Leading a **cross-functional engineering**, DevOps, and data science team to build a configuration-driven loan automation platform that determines which applications can be fully automated versus routed to analysts, preserving manual review capacity for higher-risk or complex cases.

Jr. Data Scientist

February 2024 - March 2025

Data Infrastructure

- Built an **offline model experimentation** and validation framework for credit risk decisioning, automating historical data replay, batch scoring, and champion-challenger comparisons to evaluate new model versions before production rollout.

ML Engineering

- Developed a production-grade credit risk decisioning system, building the react frontend, flask backend, model scoring APIs, azure containerized deployment, and CI/CD pipeline for underwriting model execution.
- Architected the risk model as a centralized Python package and hosted scoring API, enabling automation workflow to use the latest model version from a single source of truth.

EDUCATION

University of San Diego

MS in Applied Data Science, Data Science (GPA 3.87/4.0)

San Diego, CA

January 2022 – December 2023

University of California San Diego

Bootcamp Data Science and Visualization

San Diego, CA

August 2019 – January 2020

University of Azad (IAU)

BS in Engineering, Mechanical Engineering

Iran, Tehran

June 2013 – September 2017

PROFESSIONAL CERTIFICATES

Amazon Web Services (AWS)

AWS Certified Cloud Practitioner CLF-C01 | [Link](#)

August 2023

Microsoft Azure

Microsoft Azure AI Fundamentals AI-900 | [Link](#)

September 2023

Certificate Program

Applied Generative AI John Hopkins

May 2025 - October 2025

TECHNICAL SKILLS

- **Programming languages:** Python, SQL
- **BI / Visualization:** Tableau
- **Cloud:** AWS, Azure, Docker
- **ML Libraries:** Tensorflow, PyTorch, Scikit-learn, StatsModels
- **Developer tools:** Github, Cursor
- **Claude Code:** Skills, App design, MCP server integration, Sub-agents